

BTEC Level 3 Unit 2: Team Production Planning Sheet.

- **Project Title:** Batch Production of 5 Aluminium Spirit Levels.
- **Team Size:** 5 Members.
- **Rotational Leadership Structure:** Leadership changes at designated process milestones to ensure all members are assessed as both team members and team leaders.
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Meeting 1: Pre-Production and Workflow Strategy.

- **Timing:** Prior to workshop entry (Planning Phase).
- **My Role during this meeting:** Core Team Member / Technical Consultant.
- **Meeting Objective:** Interpret documentation, allocate initial roles, establish safety protocols, and calculate machine settings.

Planning Minutes and Decisions Made.

1. **Technical Analysis:** The team collectively analyzed Drawing Ref: SL-001. I used my advanced machining knowledge to identify that aluminium requires a high spindle speed (2500 RPM) and a 2-flute slot drill to prevent material galling and clogging.
2. **Tooling and Setup Strategy:** It was decided to utilize the **Milling Machine DRO** sub-datum memory (INC mode) to save coordinate points for the 12 mm deep vial pocket. This was planned to eliminate layout delays during the 5-unit run.
3. **Initial Role Matrix:** Member A (Dom) was appointed Initial Team Leader. I was designated as **Machining Lead** due to my high speed and precision on the lathe and mill. Member C (Cash) was assigned to Material Prep, Member D (Parmveer) to Bench Tapping, and Member E (Luke) to Quality Control.

Optimising Workflow via DRO Sub-Datums.

During our first team meeting, I explained how to optimize the milling workflow using DRO sub-datums. Manual marking-out for 5 aluminium units would cause human error and bottleneck production.

I proposed using absolute (ABS) mode for the main vice datum and saving pocket coordinates into incremental (INC) memory. Operators simply traversed until the DRO read zero, bypassing drawings between cycles. This cut cycle times by 4 minutes per part and ensured high batch repeatability.

Meeting 2: Mid-Production Review and Leadership Rotation.

- **Timing:** After completion of Unit 2 (Manufacturing Phase).
- **My Role during this meeting:** Incoming Team Leader (Handover Phase).
- **Meeting Objective:** Review production speed, evaluate peer feedback, manage a bottleneck, and rotate leadership roles.

Planning Minutes and Decisions Made.

1. **Leadership Handover:** As scheduled in our rotation plan, the leadership role officially transferred to **me** for the critical machining and assembly phase of Units 3, 4, and 5.
2. **Bottleneck Assessment:** The QC Inspector (Member E (Luke)) noted a slight dimensional drift +0.15mm on the pocket depth of Unit 2 due to tool wear.
3. **Tactical Adjustments:** Taking charge as Team Leader, you adjusted the Z-axis tool offset on the DRO to bring Unit 3 back within the strict 0.1mm tolerance. To maximize team speed, you re-allocated Member C to assist the bench-tapping station, creating a faster parallel workflow.

Upon rotating into the Team Leader role, I immediately audited our cycle times and noticed our bench work was falling behind my machining pace. While my lathe and mill cycles were highly optimised, the manual hand-tapping station created a bottleneck that stalled our fast throughput of processed aluminium bodies.

Demonstrating adaptability as a leader, I re-ordered our 5-man cell structure to ensure that deburring and tapping operated simultaneously, keeping our 5-unit target on schedule. I reallocated one team member to establish parallel bench stations, allowing filing and M6 hand-tapping to run concurrently. This tactical shift balanced our line mechanics, eliminated operator downtime, and successfully brought our total batch production time back on track.

Meeting 3: Post-Production Evaluation and Quality Sign-Off.

- **Timing:** Immediately following the completion of Unit 5 (Evaluation Phase).
- **Your Role during this meeting:** Outgoing Team Leader / Quality Sign-off.
- **Meeting Objective:** Review final batch tolerances, analyze peer feedback, and document distinction-level traits.

Planning Minutes and Decisions Made.

1. **Batch Inspection:** The team evaluated all **5 completed aluminium spirit levels**. Vials were confirmed level using a reverse-reading validation check on a surface plate, and M6 securing screws were torque-tested in the bench-tapped holes.
2. **Feedback Synthesis:** The team provided feedback on your leadership phase, praising your fast problem-solving on the DRO but noting that communication needed to be louder over the workshop machine noise.
3. **Final Accountability:** All 5 units were signed off as meeting compliance standards, with scrap and aluminium swarf successfully cleared and recycled according to workshop safety policies.

In our final meeting, I welcomed peer feedback regarding my communication style while managing the machinery. My team noted that while my machining speed and technical guidance were excellent, the high noise levels from the mill and lathe occasionally muffled my instructions. I acknowledged this constructively, recognizing that clear verbal cues are vital for workshop safety and fluid tool handovers within a 5-man team.

I documented how our forward-thinking setup of the DRO saved our team an estimated 20 minutes of layout time across the 5-unit batch, proving the commercial viability of our planning sheet. Bypassing manual height-gauge marking for each aluminium unit minimized tool downtime and eliminated layout errors. This optimized workflow proved that rigorous pre-production planning directly translates to lean, high-efficiency manufacturing in a commercial workshop environment.